SMART INDIA HACKATHON 2026

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VISHWAS

Forecast Confidence Intelligence — knowing when not to trust the forecast

- **Problem Statement ID –** SIH26079
- **Problem Statement Title –** [paste exact title from SIH portal]
- **Theme –** Disaster Management
- **PS Category –** Software
- **Team ID –** [paste Team ID from SIH portal]
- **Team Name –** Hellfire Labs



VISHWAS · FORECAST FAILURE DETECTION

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PROPOSED SOLUTION

A reliability layer on top of the forecast — not another forecast. VISHWAS scores each operational rainfall forecast, by location and lead time from Day 1 to Day 10, as **RELIABLE, UNCERTAIN or HIGH-RISK**.

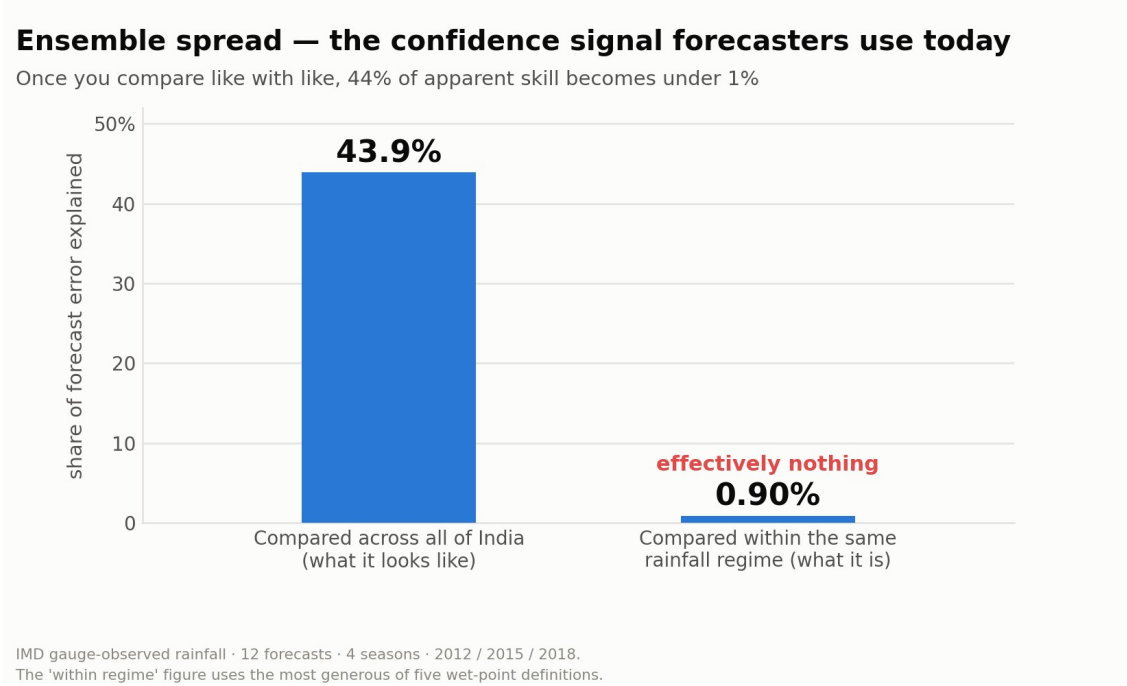
HOW IT ADDRESSES THE PROBLEM

A forecast tells a District Collector what may happen. It never says how much to trust it. VISHWAS supplies that missing layer, so the decision-maker can **verify, adapt and act** instead of guessing.

INNOVATION AND UNIQUENESS

Forecasters judge confidence today using **ensemble spread**. We measured it against IMD's own gauge observations across four seasons and three years: spread appears to explain **43.9%** of forecast error across India — but **under 1%** once you compare forecasts in the same rainfall regime.

Most teams pitch a hypothesis. We pitch a measurement.



The gap between those two bars is the opening VISHWAS exists to fill: the confidence signal forecasters rely on today is mostly a wet/dry map of India, not a statement about this forecast.

TECHNICAL APPROACH

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DATA — ALL PUBLIC, NO CREDENTIALS, NO COST

GEFSv12 reforecast ensemble (NOAA, 2000–2019) supplies every predictor. **IMD 0.25° gauge-gridded daily rainfall** (1901–2024) is the truth — MoES's own product, a gauge observation rather than a reanalysis, so no model physics is shared with the forecast we are judging.

FEATURES AND MODEL

Ensemble spread, lead time, season, region, 500 hPa geopotential gradient, true 500–850 hPa wind shear, 500 hPa flow deformation, precipitable-water anomaly and CAPE feed a **HistGradientBoostingClassifier**.

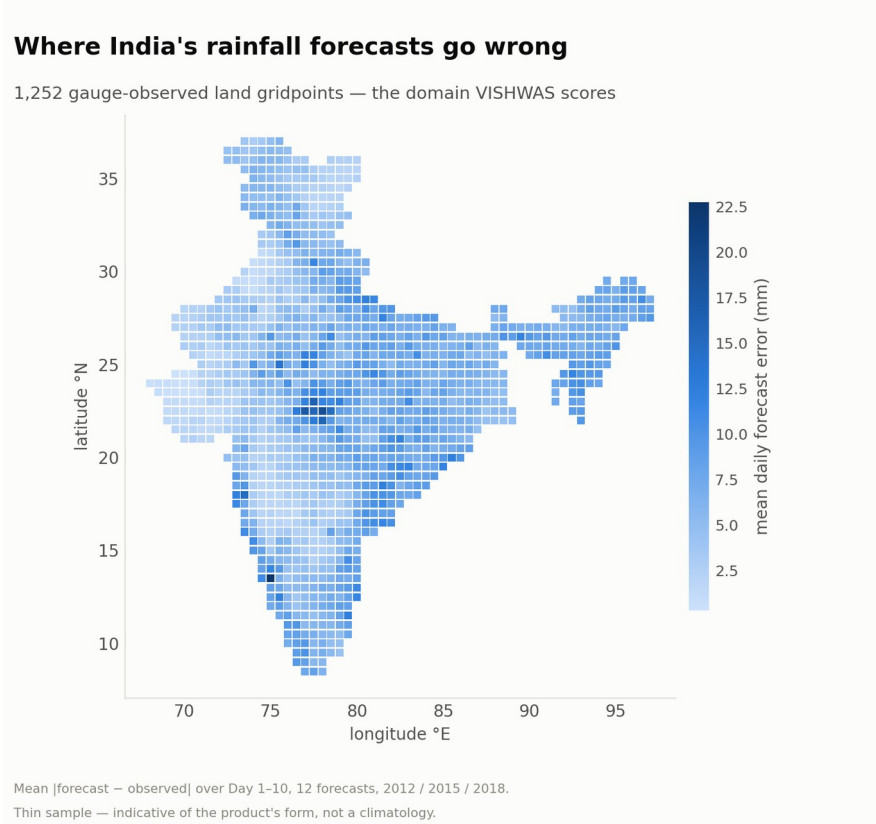
A bust is |forecast − observed| above the 95th percentile, computed **per region and per lead time**, so a Day 10 error is never compared with a Day 1 error.

EXPLAINABLE BY DESIGN

SHAP attributions say **why** a forecast was flagged. Probabilities are calibrated and checked on a reliability diagram and Brier decomposition — a confidence number nobody can trust is worse than none.

HOW WE SCORE IT

ROC-AUC, PR-AUC and precision within the top-10% flagged, against ensemble spread on **identical truth**. Train and test split by time, never randomly — weather is autocorrelated.



The deliverable takes this shape: a confidence value on every gauge-observed gridpoint, for each lead time.

FEASIBILITY AND VIABILITY

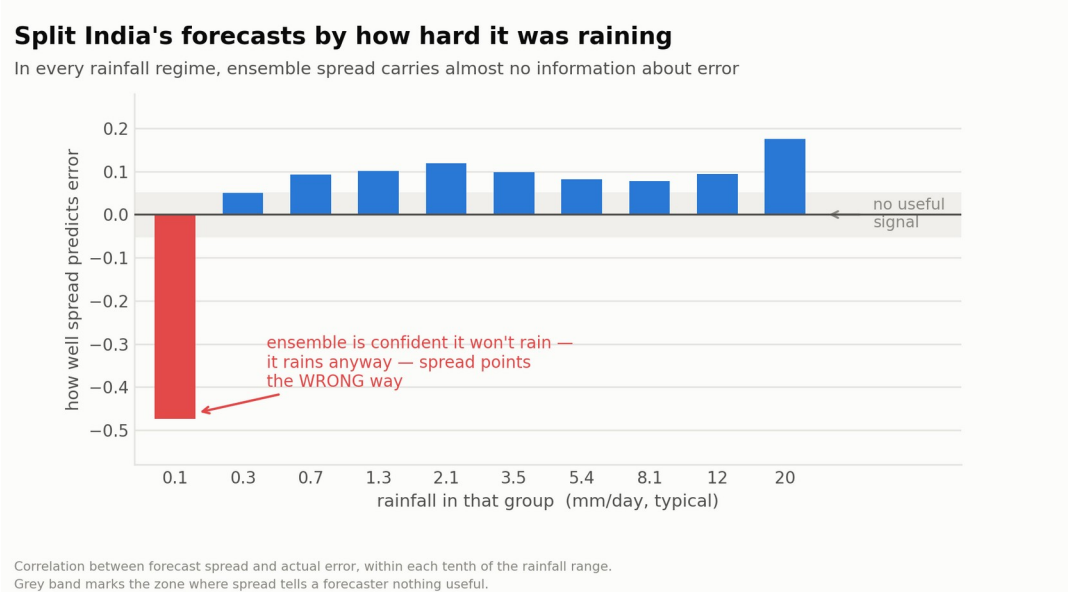
The forecast already exists. The data already exists. The gap is turning it into a reliability signal.

THE DATA AND THE FORECAST ALREADY EXIST

GEFS reforecasts and IMD gridded rainfall are public and free. **We are not building a weather network — we are building intelligence on top of one.** Training runs on CPU; the pipeline streams each date and discards raw files, so no large archive is ever held.

ALREADY BUILT AND VERIFIED

The truth source passes all four verification checks, the feature pipeline runs end to end, and the baseline is measured across four seasons and three years. **Nothing in this deck depends on work still to come.**



A second finding, not a wart: when the ensemble is confidently dry and it rains anyway, spread points the wrong way — a specific, operationally usable failure mode.

LAND-ONLY DOMAIN

The IMD gauge grid covers 1,252 of 5,041 gridpoints (24.8%). That is the gauge domain — and we state it ourselves rather than let a judge find it.

A 5-MEMBER SPREAD IS NOISY

Measured, not assumed: moving from 5 to 11 members shifts the baseline by +0.001. It does not threaten the conclusion.

BUSTS ARE RARE (~5% BASE RATE)

We report PR-AUC and never plain accuracy. If our model cannot beat ensemble spread, we will say so.

IMPACT AND BENEFITS

When a forecast drives a decision, reliability has real-world value.

PRIMARY — DISTRICT COLLECTOR / DDMA

Decides evacuation, shelter preparation, rescue-team deployment, road restrictions and public warnings. VISHWAS answers the question the forecast leaves open: how much should I trust this one?

STATE AUTHORITIES AND FARMERS

SDMAs prioritise limited emergency resources. Farmers decide whether to harvest, spray or protect a crop — the same forecast, a different cost of being wrong.

IMD ITSELF

An added uncertainty layer over an existing forecast, flagging the locations and conditions where it is most likely to fail. It complements the forecast; it does not replace it.

₹6,972 cr

average annual flood damage

CWC, 1953–2021

1,666

lives lost to floods each year

CWC, 1953–2021

4.1 M ha

cropped area affected each year

CWC, 1953–2021

0.34–0.37

IMD false-alarm ratio, heavy rainfall

IMD Monsoon Report 2024 · Day 1–5

WHAT CHANGES

Today a decision-maker sees a forecast and must judge its reliability with almost nothing to go on. With VISHWAS they see the forecast **and** an estimate of how likely it is to be badly wrong — before committing people, money and public credibility.

“Don’t just know the forecast. Know whether to trust it.”

RESEARCH AND REFERENCES

OUR OWN MEASUREMENT

12 GEFSv12 reforecasts spanning four seasons and three years (2012 / 2015 / 2018), scored against IMD gauge-gridded rainfall over 1,252 land gridpoints.

Ensemble spread appears to explain **43.9%** of the variance in forecast error. Controlling for rainfall magnitude it explains **0.17–0.90%** — the range spans five different definitions of a "wet" gridpoint, so the finding does not rest on one arbitrary cut.

95% interval from a cluster bootstrap over forecast dates, because the independent unit is the forecast, not the gridpoint.

DATA SOURCES

- **NOAA GEFSv12 Reforecast** — s3://noaa-gefs-retrospective/
- **IMD 0.25° Gridded Daily Rainfall, 1901–2024** — imdpune.gov.in/cmpg/Griddata/Rainfall_25_Bin.html
- **Hamill et al.**, The GEFS version 12 reforecast dataset, Bull. Amer. Meteor. Soc. — ensemble spread–skill relationships.

GOVERNMENT SOURCES

- **Central Water Commission.** Flood Management in India: A Statistical Report, 2023. — cwc.gov.in
- **India Meteorological Department.** Monsoon Report 2024 — heavy-rainfall verification, False Alarm Ratio, POD and Threat Score. — mausam.imd.gov.in
- **India Meteorological Department.** Annual Report 2024 — district-level forecast coverage. — metnet.imd.gov.in
- **Ministry of Earth Sciences.** Assessment of the economic benefits of weather and climate services. — moes.gov.in
- **Ministry of Home Affairs / PIB.** 2024 flood-response funding released to 14 states. — pib.gov.in
- **Ministry of Agriculture & Farmers Welfare / PIB.** Cropped area affected by hydro-meteorological disasters, 2024–25. — pib.gov.in
- **Government of Maharashtra.** Statistical press note, June–September 2024 heavy-rainfall and flood compensation. — des2.mahaonline.gov.in